

## Coding & Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 15 March 2026                   |
| <b>Team ID</b>       | xxxxxx                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

#### Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

### Solution Summary

| Field                          | Details                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>   | <b>GitHub:</b> <a href="https://github.com/sushmarangu2007-collab/credit-card-approval-prediction">https://github.com/sushmarangu2007-collab/credit-card-approval-prediction</a><br><b>Live Application:</b> <a href="https://credit-card-approval-prediction-eo1d.onrender.com/">https://credit-card-approval-prediction-eo1d.onrender.com/</a> |
| <b>Programming Language(s)</b> | Python, HTML5, CSS3                                                                                                                                                                                                                                                                                                                              |
| <b>Framework(s) Used</b>       | Flask, Scikit-learn                                                                                                                                                                                                                                                                                                                              |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Key Features Implemented</b>      | <ul style="list-style-type: none"> <li>• Credit card approval prediction using Machine Learning.</li> <li>• Data preprocessing and feature engineering.</li> <li>• Logistic Regression model training and evaluation.</li> <li>• Interactive web interface using HTML and CSS.</li> <li>• Real-time prediction through Flask.</li> <li>• Model deployment on Render.</li> <li>• Version control using GitHub.</li> </ul> |
| <b>Pending / Incomplete Features</b> | <ul style="list-style-type: none"> <li>• User authentication and login.</li> <li>• Prediction confidence score.</li> <li>• Database integration for storing application history.</li> <li>• Admin dashboard and analytics.</li> </ul>                                                                                                                                                                                    |
| <b>Setup / Run Instructions</b>      | <ol style="list-style-type: none"> <li>1. Clone the GitHub repository.</li> <li>2. Install dependencies using <code>pip install -r requirements.txt</code>.</li> <li>3. Run the application using <code>python app.py</code>.</li> <li>4. Open the browser and navigate to <code>http://127.0.0.1:5000</code>.</li> <li>5. Alternatively, access the deployed application using the Render URL.</li> </ol>               |



### Code Quality Checklist

| S.N<br>o | Criteria                                               | Status (Yes / No) |
|----------|--------------------------------------------------------|-------------------|
| 1        | Code is modular and organized into functions / classes | Yes               |
| 2        | Meaningful variable and function names are used        | Yes               |
| 3        | Code includes comments / documentation where necessary | Yes               |
| 4        | Error handling is implemented for critical operations  | Yes               |
| 5        | The application runs without critical errors           | Yes               |
| 6        | Code is committed to a version control repository      | Yes               |

#### **Additional Notes / Comments:**

- The project successfully predicts whether a credit card application is likely to be **approved or rejected** using a trained **Logistic Regression** model.
- The machine learning model was developed using **Scikit-learn** and integrated with a **Flask** web application.
- The application provides a simple and user-friendly interface for entering applicant details and viewing prediction results.
- The project is hosted on **Render** and the source code is maintained in a **GitHub** repository, making it easy to access, maintain, and extend in the future.